AI Risk Register: VoiceAssist Enterprise Deployment

Organization: Contoso Financial Services (fictional scenario) **Framework:** NIST AI 600-1 **Prepared by:** Merna Tarek, Risk Analyst **Date:** October 14, 2026

Calculated Score = Likelihood x Impact. Scoring guide: 1-4 LOW, 5-9 MEDIUM, 10-16 HIGH, 17-25 CRITICAL.

Risk ID	Category	NIST AI 600-1 Mapping	Description	Owner	Likelihood (1-5)	Impact (1-5)	Calculated Score	Status	Mitigation Notes
RISK-001	Model	Harmful Bias and Homogenization	The voice bot's intent-recognition model, trained only on urban contact-center calls, is likely to understand rural and non-native English-speaking customers less accurately, creating a risk of inequitable service outcomes for those groups.	ML Engineering Lead	4	4	**16**	Open	ML Engineering Lead to measure intent-recognition accuracy for rural and non-native English speakers, and to rerun the bias evaluation after the training data is supplemented (see RISK-007), before go-live (target date: December 31, 2026). No bias monitoring is active today, so ML Engineering Lead to set up ongoing bias monitoring.
RISK-002	Model	Confabulation	During internal testing, the voice bot gave confident answers about loan products that were updated after its last training refresh, including outdated interest rates, which could lead to customers receiving incorrect financial information.	ML Engineering Lead	5	4	**20**	Open	ML Engineering Lead to integrate Retrieval-Augmented Generation (RAG) connected to the live core banking product databases so interest rates are current, and to retest loan answers, before go-live (target date: December 31, 2026). This is a CRITICAL risk (score 20), so it is escalated to the AI Governance Board.
RISK-003	Usage	Information Security	The voice bot's LLM interface accepts open-ended natural language from external users with no input validation or adversarial prompt filtering, leaving it open to prompt injection that could expose CRM customer data or cause misuse of the bot.	Product Owner (Voice Bot)	4	4	**16**	Open	Product Owner (Voice Bot) to deploy an input guardrail and adversarial prompt filtering layer for all incoming queries, and to complete prompt injection testing, before go-live (target date: December 31, 2026).
RISK-004	Usage	Data Privacy	The voice bot handles customer financial account data and employee HR records, including health-related benefits inquiries, and formal GDPR controls and a HIPAA assessment are not yet in place, risking regulatory penalties and exposure of sensitive personal and health data.	Legal and Compliance Officer	4	5	**20**	Open	Legal and Compliance Officer to put formal data processing agreements in place, complete a privacy impact assessment, and set strict data minimization controls for health-related HR interactions, before go-live (target date: December 31, 2026). This is a CRITICAL risk (score 20), so it is escalated to the AI Governance Board.
RISK-005	Data	Data Privacy	Session data sent between the voice bot and the CRM API has not been reviewed for encryption in transit, and no data access audit has been done since the integration, risking exposure of customer PII and a breach of GDPR Article 32.	Data Steward (Customer Data)	3	4	**12**	Open	Data Steward (Customer Data) to confirm that all voice bot to CRM API traffic uses TLS 1.2 or higher, and to complete a data access audit of the integration, before go-live (target date: December 31, 2026).
RISK-006	Usage	Human-AI Configuration	The voice bot has no mandatory escalation path to a human agent for high-stakes queries such as loan application disputes or HR grievances, so customers and employees may receive unreviewed AI answers on consequential matters, risking harm from automated decisions.	Product Owner (Voice Bot)	4	4	**16**	Open	Product Owner (Voice Bot) to implement a mandatory routing rule and a smooth handoff to human customer service and HR agents for all high-stakes queries, before go-live (target date: December 31, 2026).
RISK-007	Data	Harmful Bias and Homogenization	The voice bot's training data, two years of urban call-center transcripts, contains no rural or non-native English-speaking customer interactions, which is the data-layer cause of the lower accuracy described in RISK-001.	Data Steward (Customer Data)	4	4	**16**	Open	Data Steward (Customer Data) to source, review, and ingest supplementary transcripts from rural and non-native English-speaking customers, and to document the origin of each new dataset, before go-live (target date: December 31, 2026).

Risk level of each entry

Risk ID	Score	Level
RISK-001	4 x 4 = 16	HIGH
RISK-002	5 x 4 = 20	CRITICAL
RISK-003	4 x 4 = 16	HIGH
RISK-004	4 x 5 = 20	CRITICAL
RISK-005	3 x 4 = 12	HIGH
RISK-006	4 x 4 = 16	HIGH
RISK-007	4 x 4 = 16	HIGH